

Architecture and data flow

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Requirements and service boundaries

Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Architecture and data flow. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Architecture and data flow. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Input is malformed, yet lookup runs and then a validator rejects output. Locate the first stage that should stop.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

A student request bypasses the role filter and sends teacher records directly to a generator. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A component diagram is a plan, not evidence that the running implementation enforces its arrows. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Input is malformed, yet lookup runs and then a validator rejects output. Locate the first stage that should stop.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled architecture and data flow evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of architecture and data flow. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Architecture and data flow

Four boxes for a library reply. A learner asks when the map room closes. Follow the request through the service.

Approved schedule R says 17:30. The model may phrase facts but cannot replace them.

AI ACADEMY / AI-6/2

Architecture and data flow

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Receive question

The interface forwards the map-room question.



2. Retrieve R

The application retrieves the approved time, 17:30.



3. Draft and check

A draft cites R and states 17:30; validation checks agreement before returning it.

CHANGE ONE THING

Disconnect retrieval from the application.

Explain the changed result using the matching lesson.

Label what moves along the library request, retrieval and drafting arrows. Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Four boxes for a library reply. A learner asks when the map room closes. Follow the request through the service.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A draft cites R and states 17:30; validation checks agreement before returning it.

1. Retrieve R
2. Draft and check
3. Receive question

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Draft and check A draft cites R and states

17:30; validation checks agreement before returning it. An architecture assigns responsibilities and carries data between components. Retrieval provides evidence; drafting supplies wording.

Practice 2: explain the deciding evidence

Changed-case practice: Disconnect retrieval from the application. Which conclusion is supported by the supplied facts?

1. Return review status rather than inventing a time.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Return review status rather than inventing a time. The supported-answer path lacks its required evidence input.

Repair a missing building block

Review Requirements and service boundaries (ai-6/1). Goal: Explain the mechanism and terminology of requirements and service boundaries. Short practice: Split the museum visitor's request into two operations. Which can the service perform for P, and what happens to each operation when Q has no approved card?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
